

## 12th standard Bio-Zoology Practical notes

Name: \_\_\_\_\_  
Reg.No: \_\_\_\_\_  
Subject: \_\_\_\_\_  
Class: \_\_\_\_\_ Section \_\_\_\_\_  
Date: \_\_\_\_\_ Batch \_\_\_\_\_  
Session: \_\_\_\_\_ Time \_\_\_\_\_

### Identification

| Q.No.                                                                       | Topic |
|-----------------------------------------------------------------------------|-------|
| <b>I. Experiment</b>                                                        |       |
| I.                                                                          |       |
| <b>II. Water samples / Marking of Wildlife sanctuary / Mendelian traits</b> |       |
| II.                                                                         |       |
| <b>III.A Slide</b>                                                          |       |
| III. A                                                                      |       |
| <b>IV.B Picture</b>                                                         |       |
| IV. B                                                                       |       |
| <b>V.C Genetics</b>                                                         |       |
| V. C                                                                        |       |

| Q. No.       | Ex.No. | Content                                                               |
|--------------|--------|-----------------------------------------------------------------------|
| <b>I</b>     |        | <b>Experiments</b>                                                    |
|              | 1.     | A Fermentation by yeast                                               |
|              | 2.     | B Determination of colour and pH in the given water samples           |
| <b>II</b>    |        | <b>Photographs/Charts</b>                                             |
|              | 3.     | <b>Marking of wildlife sanctuary and National parks in Indian map</b> |
|              |        | A1 Kaziranga National Park                                            |
|              |        | A2 Point Calimere wildlife and Bird Sanctuary                         |
|              |        | A3 Gir National Park and wildlife Sanctuary                           |
|              |        | A4 Periyar Wildlife Sanctuary                                         |
|              |        | A5 Mudumalai Wildlife Sanctuary and National Park                     |
|              | 4.     | B Human Mendalian traits                                              |
|              | 5.     | C ABO blood grouping                                                  |
| <b>III A</b> |        | <b>Slides</b>                                                         |
|              | 6.     | A1 Human sperm                                                        |
|              | 7.     | A2 Human ovum                                                         |
|              | 8.     | A3 Paramecium - Conjugation                                           |
|              | 9.     | A4 Entamoeba histolytica                                              |
|              | 10.    | A5 Thymus gland – T.S                                                 |
|              | 11.    | A6 Lymph node – T.S                                                   |
| <b>IV B</b>  |        | <b>Pictures</b>                                                       |
|              | 12.    | B1 Structure of t-RNA                                                 |
|              | 13.    | B2 Homologous organs                                                  |
|              | 14.    | B3 Analogous organs                                                   |
|              | 15.    | B4 Animal cloning –Dolly                                              |
|              | 16.    | B5 Human Insulin production – Flowchart                               |
| <b>V C</b>   |        | <b>Genetics</b>                                                       |
|              | 17.    | C1 Normal Human Karyotype                                             |
|              | 18.    | C2 Autosomal Anomaly – Patau’s Syndrome                               |
|              | 19.    | C3 Sex Chromosomal Anomaly – Turner’s Syndrome                        |
|              | 20.    | C4 X-Linked Disease - Haemophilia                                     |
|              | 21.    | C5 Autosomal Disease – Sickle cell anemia                             |

## **Bio-Zoology Practical Question Paper**

**Time: 1:30 hr**

**Marks: 10**

- I. Analyse the given samples I, II and III for fermentation process. Write the aim, principle, procedure and inference of the experiment. (Procedure - 1, Experiment - 1, Result –  $\frac{1}{2}$ ) **2½ marks**
- II. Analyse the given water samples I, II and III for colour and pH. Tabulate your results and find out which water is suitable for consumption. (Procedure - 1, Experiment -  $\frac{1}{2}$ , Result –  $\frac{1}{2}$ )  
(OR)  
Mark the location of the given wildlife Sanctuary and National Parks in India map. Add a note on its location and significance. (Location – 1, Significance – 1)  
(OR)  
Mention any 4 Mendelian traits in your body and write their phenotype and genotype.  
(Phenotype – 1, Genotype – 1) **2marks**
- III. Identify the given slide 'A'. Write any two diagnostic features with diagram.  
(Diagram –  $\frac{1}{2}$ , Any two diagnostic features –  $\frac{1}{2}$ ) **1mark**
- IV. Identify the given picture 'B'. Write any two comments.  
(Identification –  $\frac{1}{2}$ , Any two comments –  $\frac{1}{2}$ ) **1mark**
- V. Identify the chromosomal abnormality in 'C'. Write any two characteristic features.  
(OR)  
Analyse and identify the genetic disease in pedigree chart given in 'C'. Answer the given questions.  
(Identification –  $\frac{1}{2}$ , Characteristic features / Answers –  $\frac{1}{2}$ ) **1mark**

**Total: 7½ mark**  
**Record: 1½ mark**  
**Skil: 1 mark**  
**Maximum marks: 10**

## I - Experiments

### Exercise No: 1. Fermentation by yeast

**Aim:** To demonstrate the process of fermentation by yeast.

**Requirements:**

1. Glucose solution    2. Raw sugar solution    3. Jaggery with salt solution    4. yeast granules

**Principle:**

Sugar is converted into ethyl alcohol and CO<sub>2</sub>. It occurs in yeast and bacteria.

**Procedure:**

1. Take 2 ml of the given samples I, II and III in three clean test tube respectively.
2. Add few granules of yeast in all the test tubes and plug the tubes with cotton wool.
3. Wait while fermentations take place and note that time taken.
4. Appearance of effervescence in the test tube indicates that fermentations has taken place.
5. Remove the Cotton wool and pass the gas into a test tube containing lime water.
6. The lime water turns milky indicating that the gas evolved during fermentation is carbon dioxide.

| S.No. | Sample | Time taken                          | Inference             |
|-------|--------|-------------------------------------|-----------------------|
| 1.    | I      | Appearance of effervescence quickly | Simple sugar present  |
| 2.    | II     | Appearance of effervescence slowly  | Complex sugar present |
| 3.    | II     | No appearance                       | No fermentation       |

**Result:**

Yeast has an enzyme which catalyses the fermentation process.

1. To given a sample I is Simple sugar present, it is Fermentation quickly.
2. To given a sample II is Complex sugar present, it is Fermentation slowly.
3. To given a sample III is No Appearance, it is no Fermentation.

## II- Water Samples

### Exercise No: 2. Determination of colour and pH in the given water samples

**Aim:** To investigate the colour and pH in the given water samples and determining the quality of water for consumption.

**Requirements:**

1. pH paper and colour chart    2. Water samples    3. Dropper    4. Test tubes

**Principle:**

Solutions with a pH between 0-7 are acidic, pH between 7-14 are basic and pH 7 is considered neutral.

**Procedure:**

1. Observe and tabulate the colour of the water samples I, II, and III taken in test tube against a white background
2. Take three different water solution in separate test tube and label them
3. A piece of pH paper is dipped into the sample and compared with that of the colour in the pH chart.

| S.No. | Sample | Time taken                 | Inference |
|-------|--------|----------------------------|-----------|
| 1.    | I      | Colour less (Rain water)   | 7         |
| 2.    | II     | Green colour (River water) | 6.5       |
| 3.    | II     | Brown colour (Pond water)  | 8         |

**Result:**

1. To given a sample I is found to be pH 7 is neutral in nature, it is fit for consumption.
2. To given a sample II is found to be pH 6.5 is acidic in nature, it is not suitable for consumption
3. To given a sample III is found to be pH 8 is alkaline nature, it is not suitable for consumption.



## II - Photographs/Charts

### Marking of wildlife sanctuary and National parks in Indian map

#### Exercise No: 3 (A1): Kaziranga National Park

**Identification:** The given Indian map is identified as Kaziranga National Park.

**Location :** Nagaon district, Assam

**Significance:**

- 1.Kaziranga National Park 430 Square kilometres.
- 2.More than 2200 Indian one horned rhinoceros in Kaziranga National park.
- 3.Kaziranga was declared as Tiger Reserve in 2006.

#### Exercise No: 3 (A2): Point calimere wildlife and bird sanctuary

**Identification:** The given Indian map is identified as Point calimere wildlife and bird sanctuary.

**Location:** point calimere – kodiyaikarai, Nagapattinam district.

**Significance:**

It was created for the conservation of near threatened species, Black buck antelope and endemic mammal species of India.

#### Exercise No: 3 (A3): Gir National park and wildlife sanctuary

**Identification:** The given Indian map is identified as Gir National park and wildlife sanctuary.

**Location:** Gir, Gujarat.

**Significance:**

- 1.Gir is the only natural habitat world popular Asiatic lion.
- 2.It covers total area of 1412 square kilometres.
- 3.The jackal, striped Hyena and Indian Fox are some of the smaller carnivores found in Gir Forest.

#### Exercise No: 3 (A4): Periyar wildlife sanctuary

**Identification:** The given Indian map is identified as Periyar wildlife sanctuary.

**Location:** Kerala

**Significance:**

- 1.Apart from elephants, the other animals to be seen in the Periyar Sanctuary are gaur, wild bigs, sambar, Mouse deer,etc.,
- 2.There are now an estimated 40 tigers here.

#### Exercise No: 3 (A5): Mudumalai wildlife sanctuary and national park

**Identification:** The given Indian map is identified as Mudumalai wildlife sanctuary and national park.

**Location:** Nilagiri hills, Nilgiri District, Tamilnadu.

**Significance:**

- 1.The protected area is home to several endangered and vulnerable species including Indian elephants Bengal tigers.
- 2.There are at least 266 species of birds in the sanctuary including critically endangered Indian white rumped vulture and long billed vulture.

### Exercise No: 4 (B): Human Mendelian traits

| Traits            | Possible alleles              | Your phenotype | No | %     | Your genotype | No | %     |
|-------------------|-------------------------------|----------------|----|-------|---------------|----|-------|
| Hair curl         | Curly (H)<br>Straight (h)     | Straight (h)   | 14 | 93.3% | Recessive (h) | 14 | 93.3% |
| Tongue rolling    | Roller (T)<br>Non roller (t)  | Roller (T)     | 6  | 40%   | Dominant (T)  | 6  | 40%   |
| Earlobes          | Free lobe (F)<br>Attached (f) | Attached (f)   | 13 | 86.6% | Recessive (f) | 13 | 86.6% |
| Handedness        | Right (R)<br>Left (L)         | Right (R)      | 15 | 100%  | Dominant (R)  | 15 | 100%  |
| Shape of the face | Oval (O)<br>Square (o)        | Square (o)     | 9  | 60%   | Recessive (o) | 9  | 60%   |

**Result:**

1. Did you have mostly dominant or recessive traits? - Recessive.
2. For which trait were most students dominant? - Right handness (R) 100%.
3. For which trait were most students recessive? - Stright hair (d) 93.3%.

**Exercise No: 5 (C): ABO blood grouping**

| 1.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| +                    | -      | +                     | A+          |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be A+ group

| 2.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| +                    | -      | -                     | A-          |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be A- group

| 3.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| -                    | +      | +                     | B+          |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be B+ group

| 4.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| -                    | +      | -                     | B-          |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be B- group

| 5.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| +                    | +      | +                     | AB+         |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be AB+ group

| 6.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| +                    | +      | -                     | AB-         |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be AB- group

| 7.Agglutination with |        |                       | Blood Group |
|----------------------|--------|-----------------------|-------------|
| Anti A               | Anti B | Anti D                |             |
| -                    | -      | +                     | O+          |
| (+) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be O+ group

| 8. Agglutination with |        |                       | Blood Group |
|-----------------------|--------|-----------------------|-------------|
| Anti A                | Anti B | Anti D                |             |
| -                     | -      | -                     | O-          |
| (+ ) Agglutination    |        | (-) Non Agglutination |             |

**Result:** The given blood is found to be O- group

### III. A - Slides

#### Exercise No: 6 (A1): Human sperm

**Identification:** The given slides is identified as human sperm

**Comments:**

1. The human sperm is microscopy flagellated and a motile male gamete.
2. The sperm is composed of a head, neck, middle piece and tail.

#### Exercise No: 7 (A2): Human ovum

**Identification:** The given slide is identified as human ovum

**Comments:**

1. human ovum is non cleidoic and a alecithal female gamete.
2. The ovum is surrounded by three covering names vitelline membrane, zona pellucida and Corona radiata.

#### Exercise No: 8 (A3): Paramecium - Conjugation

**Identification:** The given slides is identified as Paramecium conjugation

**Comments:**

1. Conjugation is a form of sexual reproduction, where in two individuals called conjugation.
2. Mutually exchange s nuclear materials and then get separated.

#### Exercise No: 9 (A4): Entamoeba histolytica

**Identification:** The given slide is identified as Entamoeba histolytica.

**Comments:**

1. It is a endoparasitic protozoan.
2. Which causes amoebiasis are amoebic dysentery.

#### Exercise No: 10 (A5): Thymus gland – T.S

**Identification:** The given slide is identified as thymus gland - T.S

**Comments:**

1. Thymus is a primary Lymphoid bilobed organ
2. It is located behind the sternum and above the heart.

#### Exercise No: 11 (A6): Lymph node – T.S

**Identification:** The given slide is identified as lymph node – T.S

**Comments:**

1. Lymph node is a small bean shaped structure.
2. Lymph node has three zones: cortex, paracortex and medulla.

### IV.B – Pictures

#### Exercise No: 12 (B1): t-RNA Structure

**Identification:** The given model is identified as t-RNA Structure.

**Comments:**

1. t-RNA formerly referred to as soluble RNA
2. It is a type of RNA with clover leaf structure.

#### Exercise No: 13 (B2): Homologous organs

**Identification:** The given picture is identified as Homologous organs.

**Comments:**

1. Structures which are similar in origin but perform different function are called Homologous structure.
2. Example - four Limbs of terrestrial vertebrates birds and bat etc.,

**Exercise No: 14 (B3): Analogous organs**

**Identification:** The given picture is identified as Analogous organs.

**Comments:**

1. Organism having different structural patterns but the similar function is termed as analogous structure
2. Example - Wings of bird and insect.

**Exercise No: 15 (B4): Animal cloning - dolly**

**Identification:** The given picture is identified as Animal cloning – dolly.

**Comments:**

1. Cloning is the process to produce genetically identical individuals of an organism their naturally or artificially.
2. Dolly was first mammal clone developed by willmott and campbell in 1997.

**Exercise No: 16 (B5): Human insulin production**

**Identification:** The given picture is identified as human insulin production.

**Comments:**

1. Production of insulin by Recombinant DNA technology started in the late 1970s.
2. This techniques involved the Insertion of human insulin gene on the plasmids of E.coli.

**V.C – Genetics****Exercise No: 17 (C1): Normal Human Karyotype**

**Identification:** The given picture is identified as Normal Human Karyotype.

**Comments:**

1. Karyotyping is a technique through which a complete set of chromosomes are separated from here cell and are arranged in pairs.
2. A diagrammatic representation of chromosomes is called ideogram.

**Exercise No: 18 (C2): Autosomal Anomaly - Patau's syndrome**

**Identification:** The given picture is identified as patau's syndrome.

**Comments:**

1. It is one of the autosomal aneuploids formed due to trisomic condition of chromosome 13.
2. It is caused by meiotic non- disjunction of chromosomes.

**Exercise No: 19 (C3): Sex Chromosomal Anomaly - Turner's syndrome**

**Identification:** The given picture is identified as Turner's syndrome.

**Comments:**

1. This genetic disorder is due to the loss of an x chromosome resulting in a karyotype of  $44+XO=45$ .
2. It is caused due to meiotic non- disjunction of allosomes.

**Exercise No: 20 (C4): X-Linked Disease – Haemophilia**

**Identification:** The given pedigree chart is identified as X-Linked Disease – Haemophilia

**Discussion Questions:**

1. Observe the given pedigree chart and identify the affected individuals and carriers in the II generation.  
**Ans:** The affected individuals are Male and II generation carriers are Female.
2. Why are men affected often in X-linked inheritance?  
**Ans:** It is caused by a recessive X-linked gene. Hence, mens are affected.
3. What is the pattern of inheritance in the given pedigree chart?  
**Ans:** Criss – Cross or Zig – Zag pattern of inheritance.
4. Why are women said to be carriers in X-linked inheritance?  
**Ans:** In homozygous condition recessive alleles are not expressed the disease. Hence, womens are considered as carriers.
5. How is Heamophilia caused?  
**Ans:** Heamophilia is caused by a recessive X-linked gene.

### Exercise No: 21 (C5): Autosomal Disease – Sickle cell anemia

**Identification:** The given pedigree chart is identified as Autosomal Disease – Sickle cell anemia.

#### Discussion Questions:

1. Observe the given pedigree chart and give reasons for the occurrence of the disease in the second generation.

**Ans:** It is an autosomal linked recessive trait that can be transmitted from parents to the offspring. Hence, The second generation is affected by the disease.

2. Will males and females be equally affected in this type of inheritance? Give reasons.

**Ans:** Yes, It is an autosomal disease. The recessive characters are transmitted through body chromosomes. So, male and females are equally affected.

3. What is the genotype of the disease and carrier of sickle cell individual?

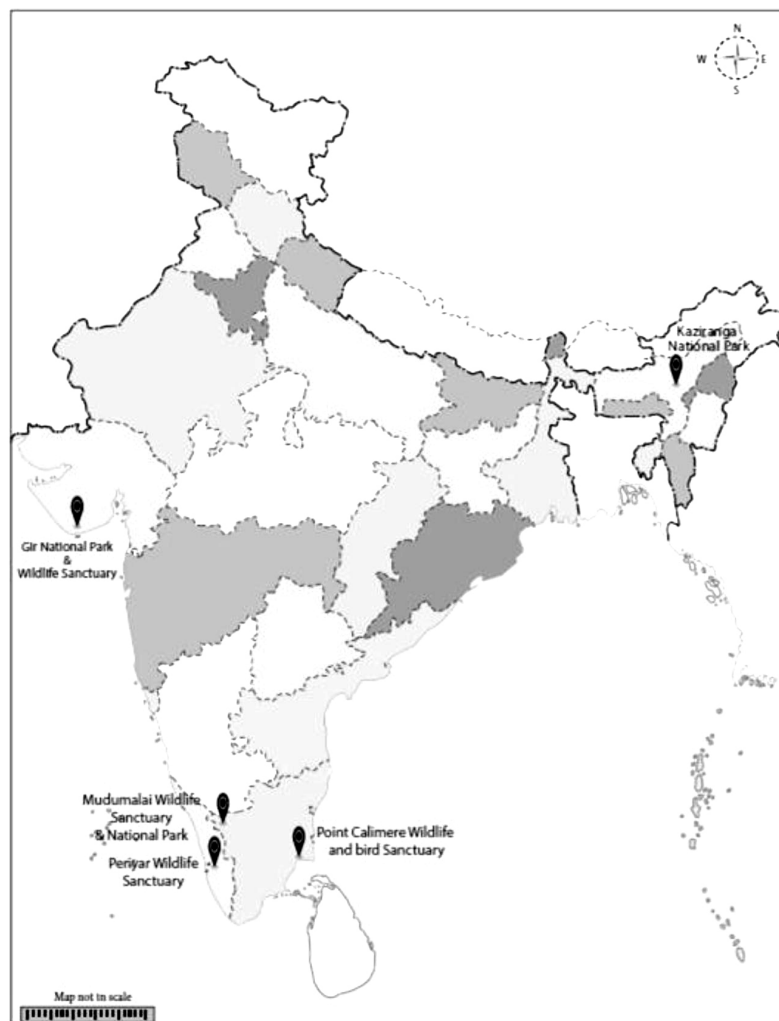
**Ans:**  $Hb^S Hb^S$  is the genotype of the disease and  $Hb^A Hb^S$  is the sickle cell carrier.

4. How is sickle cell anemia disease caused?

**Ans:** Sickle cell anemia is caused by the substitution of Glutamic acid by valine at the sixth position of the beta globin chain of the haemoglobin molecules.

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### Marking of wildlife sanctuary and National parks in Indian map

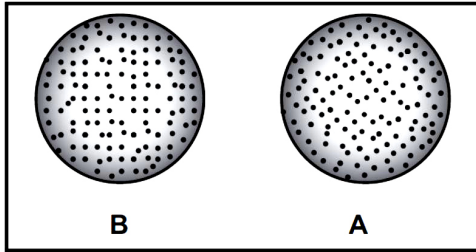


## Human Mendalian traits

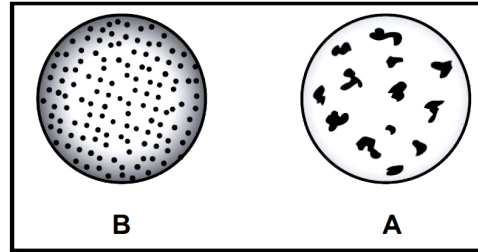
|                         | DOMINANT                                                                                              | RECESSIVE                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Cleft chin              |  Have cleft          |  No cleft             |
| Hair curl               |  Curly               |  Straight             |
| Tongue rolling          |  Roller              |  Non roller           |
| Dimples                 |  Dimple              |  No dimple            |
| Ear lobes               |  Free lobe          |  Attached lobe       |
| Interlocking fingers    |  Left thumb on top |  Right thumb on top |
| Handedness              |  Right             |  Left               |
| Widow's peak            |  Widow's peak      |  Straight           |
| Shape of face           |  Oval              |  Square             |
| Finger mid digital hair |  Hair              |  No hair            |

## ABO blood grouping

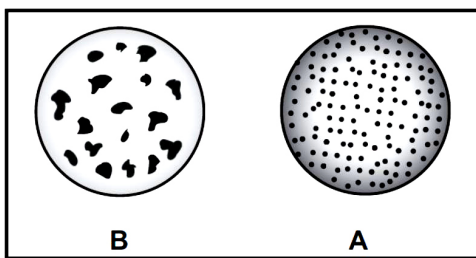
Group O



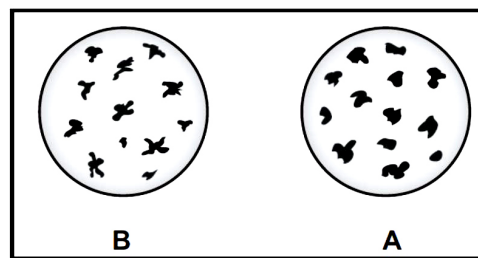
Group A



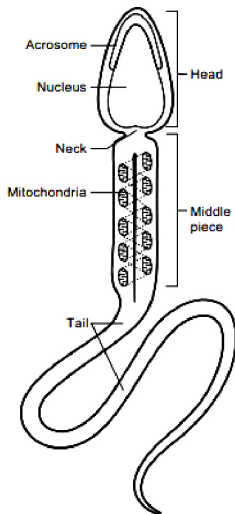
Group B



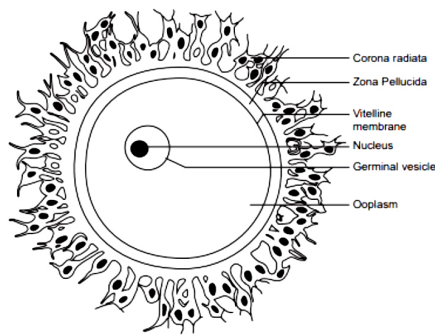
Group AB



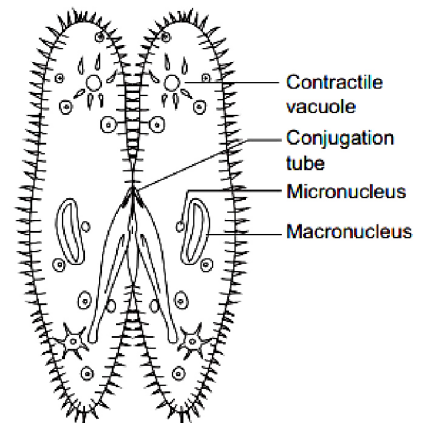
Human sperm



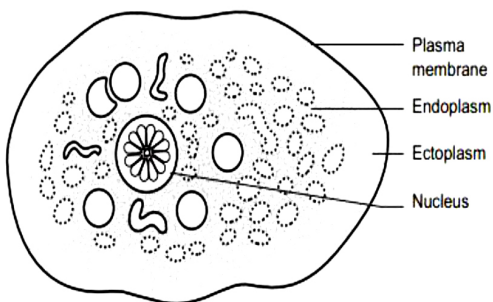
Human ovum



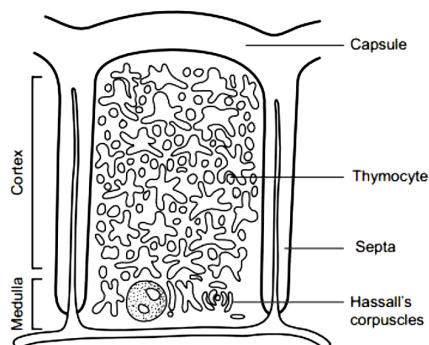
Paramecium - Conjugation



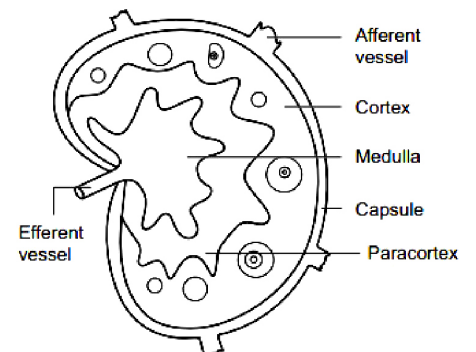
Entamoeba histolytica



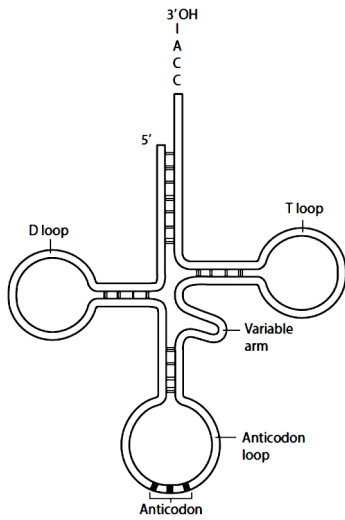
Thymus gland – T.S



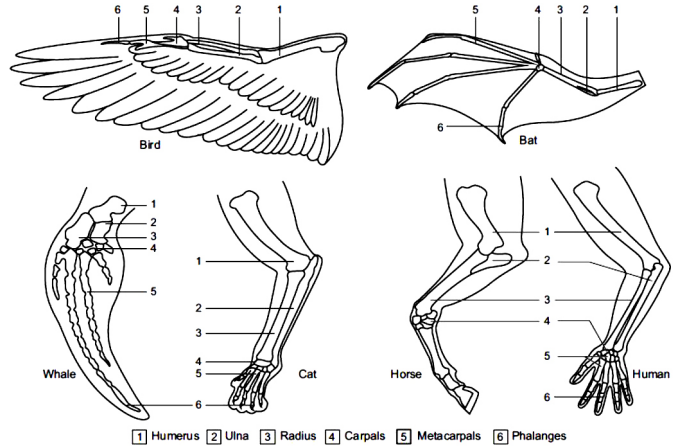
Lymph node – T.S



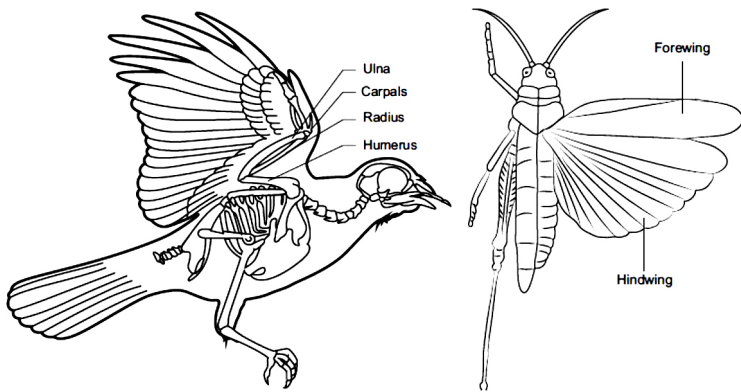
## t-RNA Structure



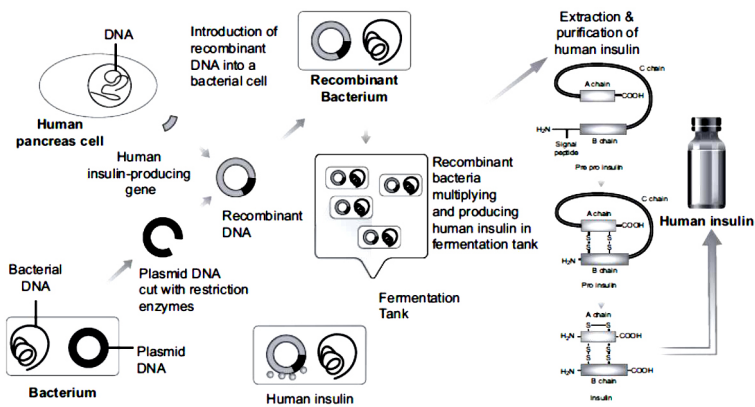
## Homologous organs



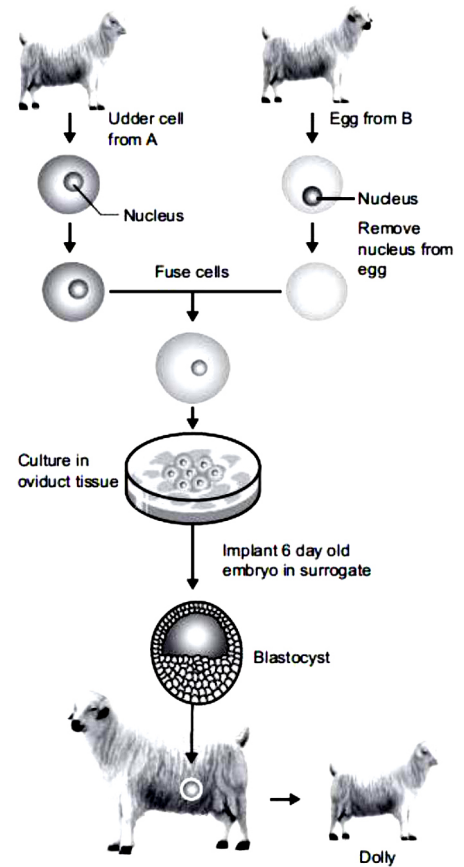
## Analogous organs



## Human insulin production

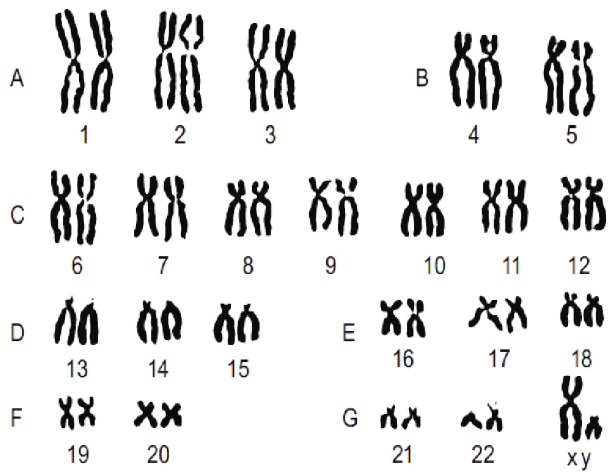


## Animal cloning – dolly

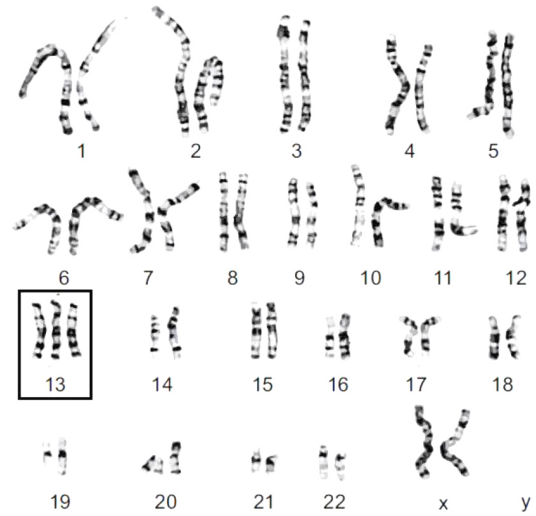




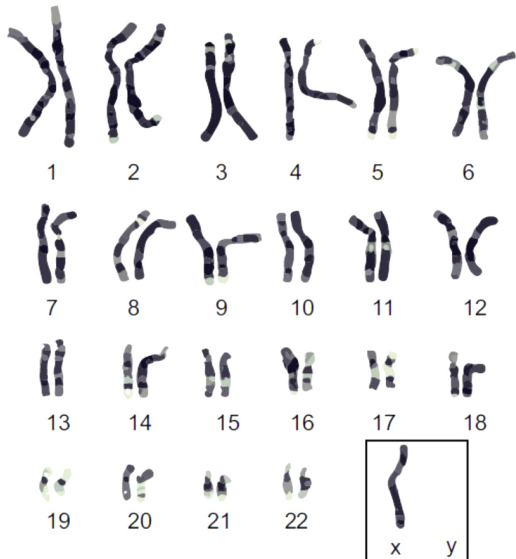
### Normal Human Karyotype



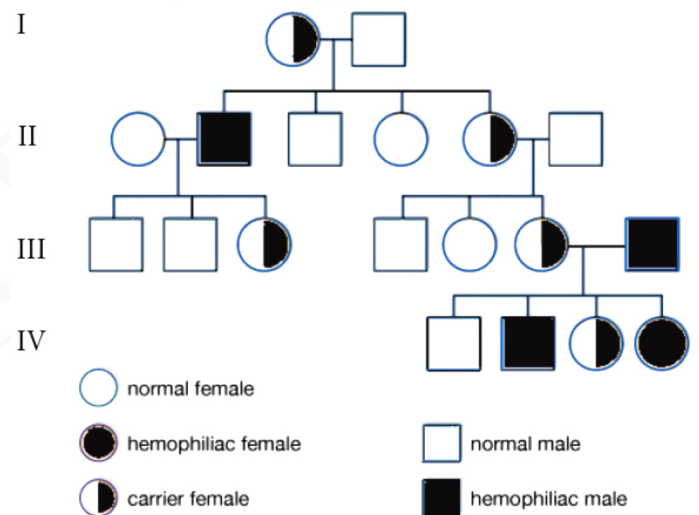
### Autosomal Anomaly - Patau's syndrome



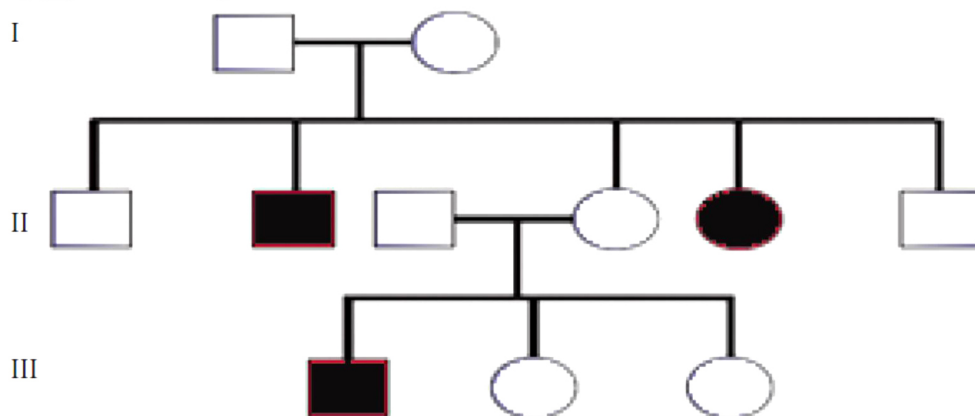
### Sex Chromosomal Anomaly - Turner's syndrome



### X-Linked Disease – Haemophilia



### Autosomal Disease – Sickle cell anemia



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